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The Chandrayaan-1 mission is aimed at high-resolution remote sensing of the moon in visible, near-infrared, low-energy X-ray and high-energy X-ray regions.

At first, Chandrayaan-1 reached a highly elliptical orbit. After encircling the earth for a while, the spacecraft was taken into two more elliptical orbits whose apogees were still higher at 37,000 km and 73,000 km, respectively. This all was done at a very precise moment by firing the spacecraft's liquid apogee motor (LAM) when the spacecraft was near perigee. Subsequently, the LAM was fired to take the spacecraft to a high orbit whose apogee lied at about 387,000 km. When the Chandrayaan-1 reached the vicinity of the moon, the spacecraft was oriented in a particular way and its LAM was fired again to slow down the spacecraft sufficiently to enable the gravity of the moon to acquire it into an elliptical orbit.

About 20 days from the date of launch, Chandrayaan-1 was in the required moon orbit. When the orbital height of Chandrayaan-1 was lowered to its intended 100km height from the lunar surface, the MIP was ejected from Chandrayaan-1 at the earliest on to the lunar surface in a chosen area.

The spacecraft

The spacecraft for lunar mission is cuboid in shape with each side approximately 1.50 metres. It accommodates eleven science payloads. The 3-axis stabilised spacecraft uses two star sensors, gyros and four reaction wheels.

A canted single-sided solar array will provide the required power during all phases of the mission. This deployable solar array consisting of a single panel generates 700W of peak power. The solar array along with yoke was stowed on the south deck of the spacecraft in the launch phase. During eclipse, the spacecraft was powered by lithium-ion (Li-ion) batteries. After deployment, the solar panel plane was canted by 30° to the spacecraft pitch axis.

The spacecraft employs an X-band, 0.7m diameter parabolic antenna for payload data transmission. The antenna employs a dual gimbal mechanism to track the earth station when the spacecraft is in lunar orbit.

The spacecraft uses a bipropellant integrated propulsion system to reach the lunar orbit as well as orbit and attitude maintenance while orbiting the moon. The propulsion system carries the required propellant for a mission life of two years, with adequate margin.

The telemetry, tracking and command communication is in S-band frequency. The scientific payload data transmission is in X-band frequency.

The spacecraft has three solidstate recorders (SSRs) on board to record data from various payloads. SSR-1 will store science payload data and has capability of storing 32GB data. SSR-2 will store science payload data along with spacecraft attitude information (gyro and star sensor), satellite house-keeping and other auxiliary data. The storage capacity of SSR-2 is 8 GB. MB (moon mineralogy mapper) payload has an independent SSR with 10GB capacity.

Scientific payloads

The indigenously developed payloads are:

1. Terrain-mapping stereo camera in the panchromatic band, having 5m spatial resolution and 20km swath
2. Hyper-spectral imaging camera operating in the 400-950nm band with a spectral resolution of 15 nm and spatial resolution of 80 metres with a swath of 20 km
3. Lunar laser-ranging instrument with height resolution of about 10 metres
4. High-energy X-ray spectrometer using cadmium-zinc-telluride